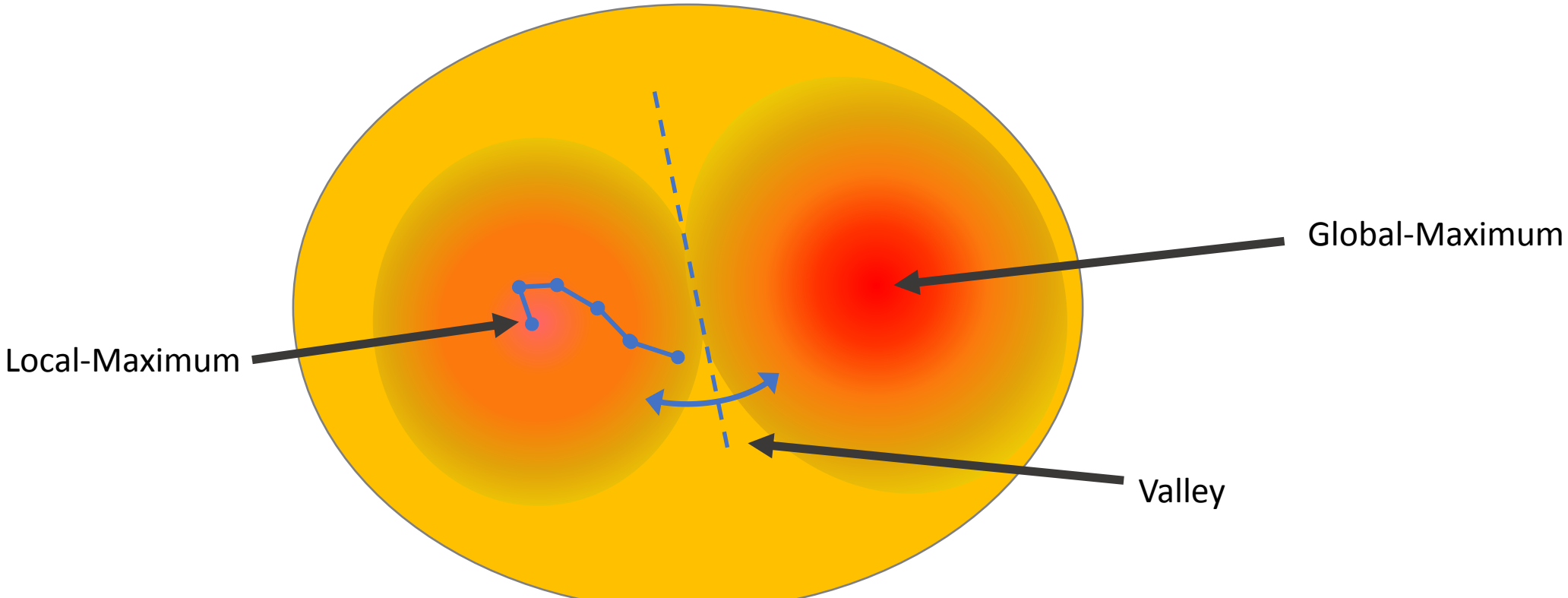
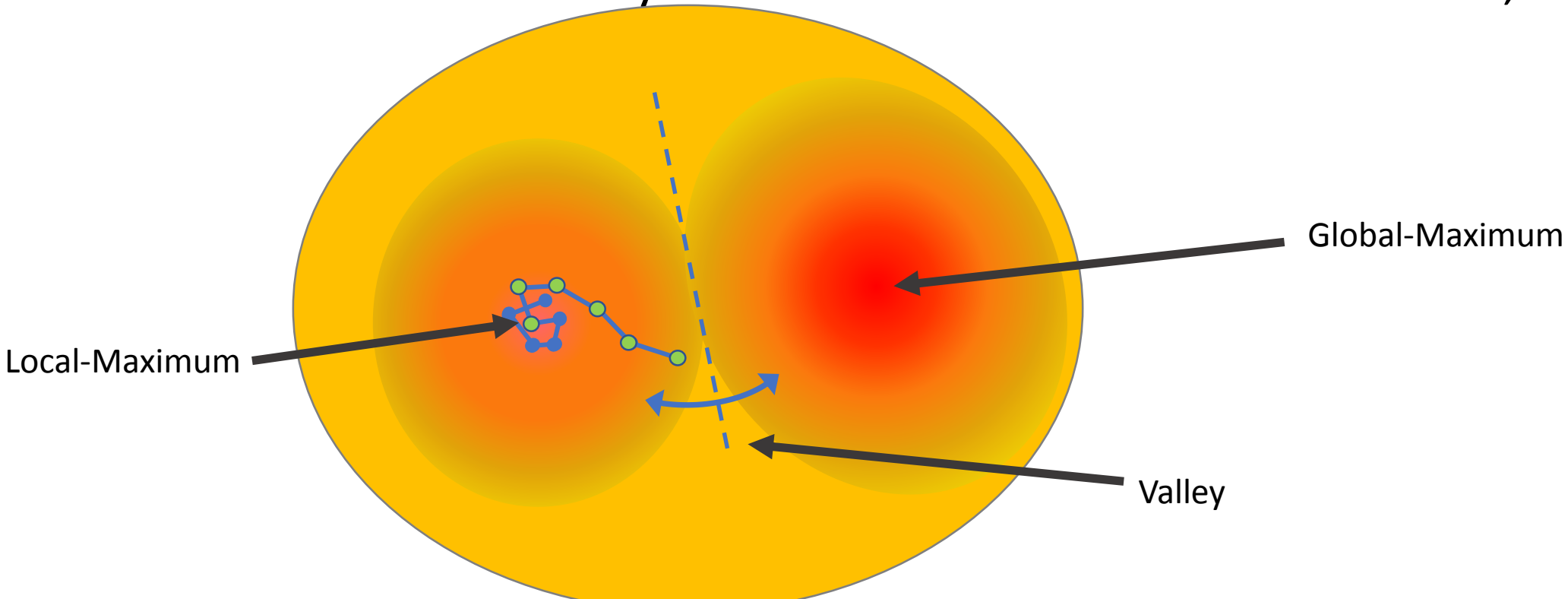
Quick Recap: Hill-Climbing:

- **Idea:** Always follow the best neighbor
- **Pros:** Easy to implement; performs well in convex spaces
- **Cons:** Gets stuck in local-maximum; lost in plateaus; ridges



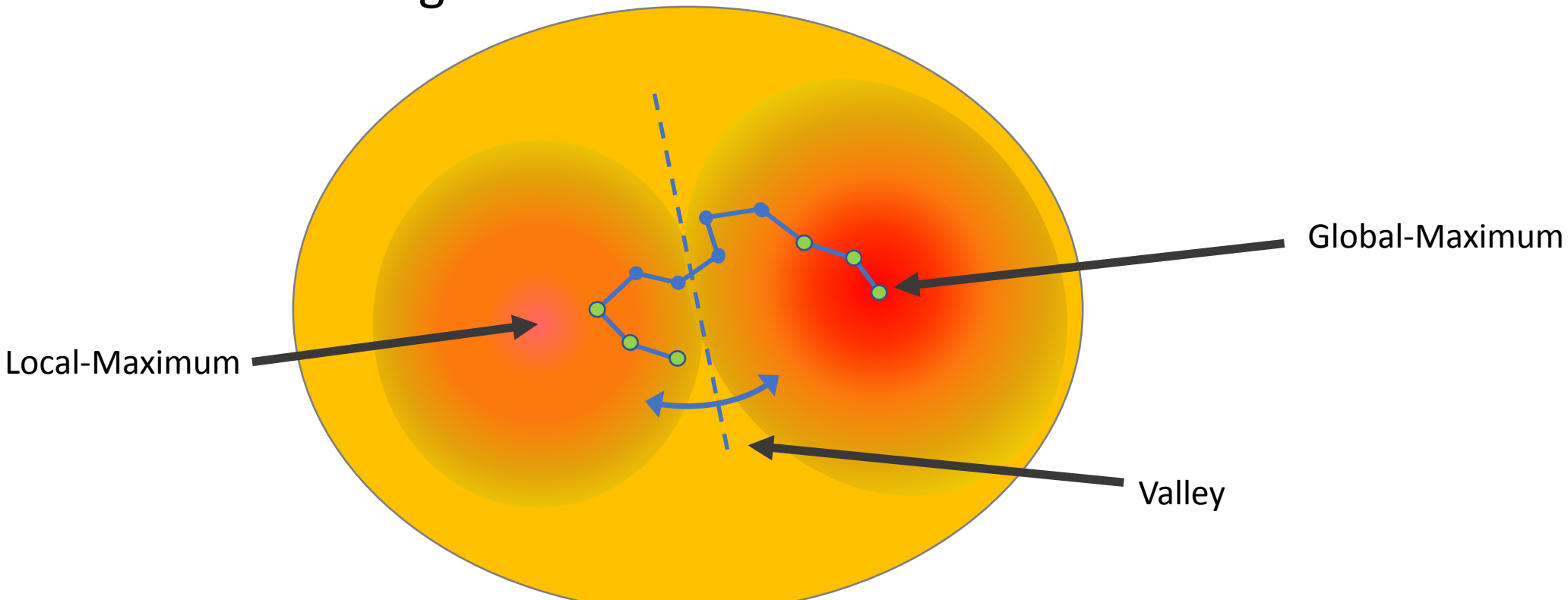
Quick Recap: Beam-Search

- **Idea:** Keep track of N-th good-looking neighbors; expand them to see if they lead somewhere better.
- **Pros:** Has “memory”, can backtrack a little to overcome HC limitation
- **Cons:** Ability to overcome local-maxima is limited; Memory.



Simulated Annealing

- **Idea:**
  - Instead of following the best neighbor(s), sometimes expand less-optimal neighbors
  - May lead to global-maxima that lie beyond poor-looking regions



Simulated Annealing

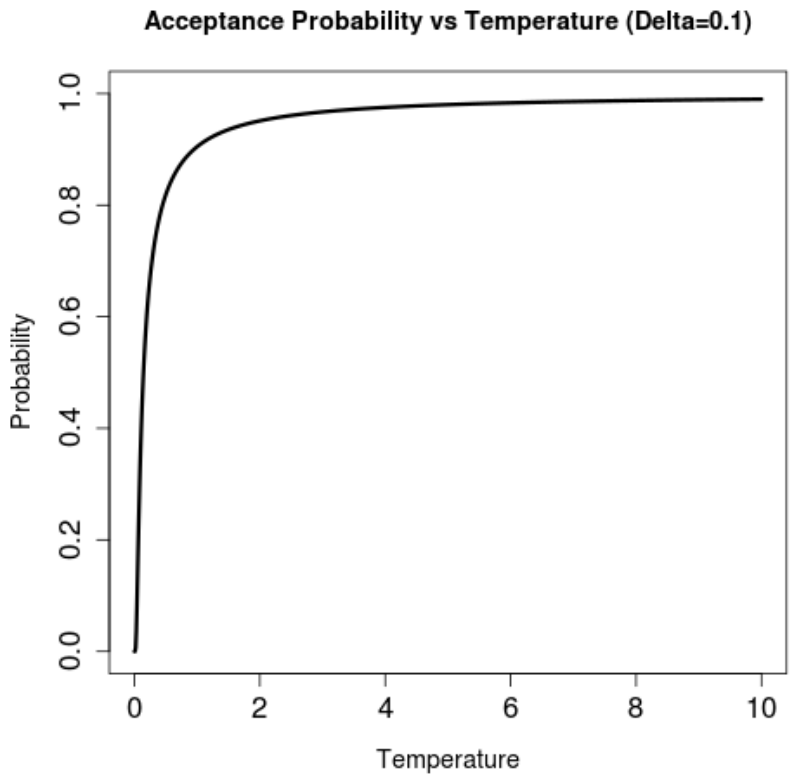
- **How do you choose when to visit neighbors?**
  - Always accept better ones
  - Accept worse ones probabilistically:
  - Borrow from physics / metallurgy (hence name):
    - Higher “temperature” = more volatile system.
    - Lower “temperatures” = more stable system.

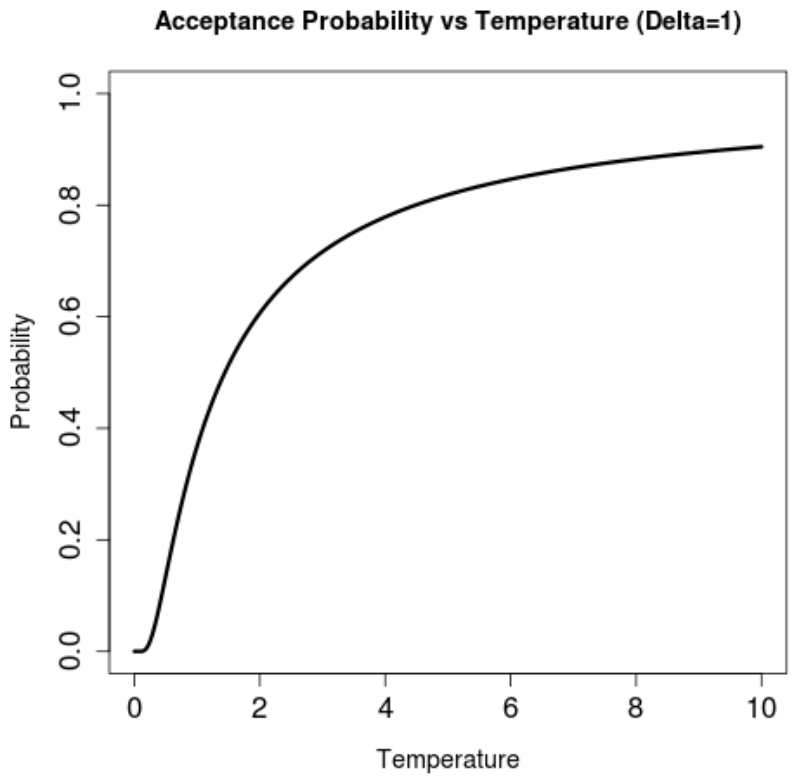
$$P = exp\left(\frac{\Delta E}{\mathrm{T}}\right)$$

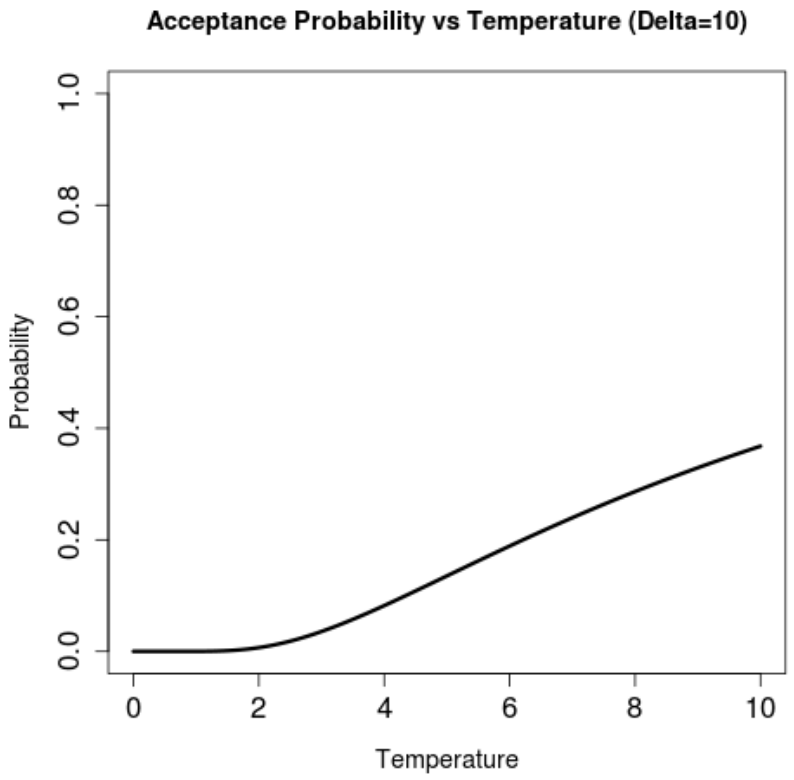
- ΔE the difference in value/cost between states
- T is a parameter of the algorithm:
  - Values and rate of change determined by user

Simulated Annealing

- **Acceptance of new state is a function of temperature and how good/bad the state is compared to current one:**







- Relatively good states (low-delta) are almost always accepted

Simulated Annealing

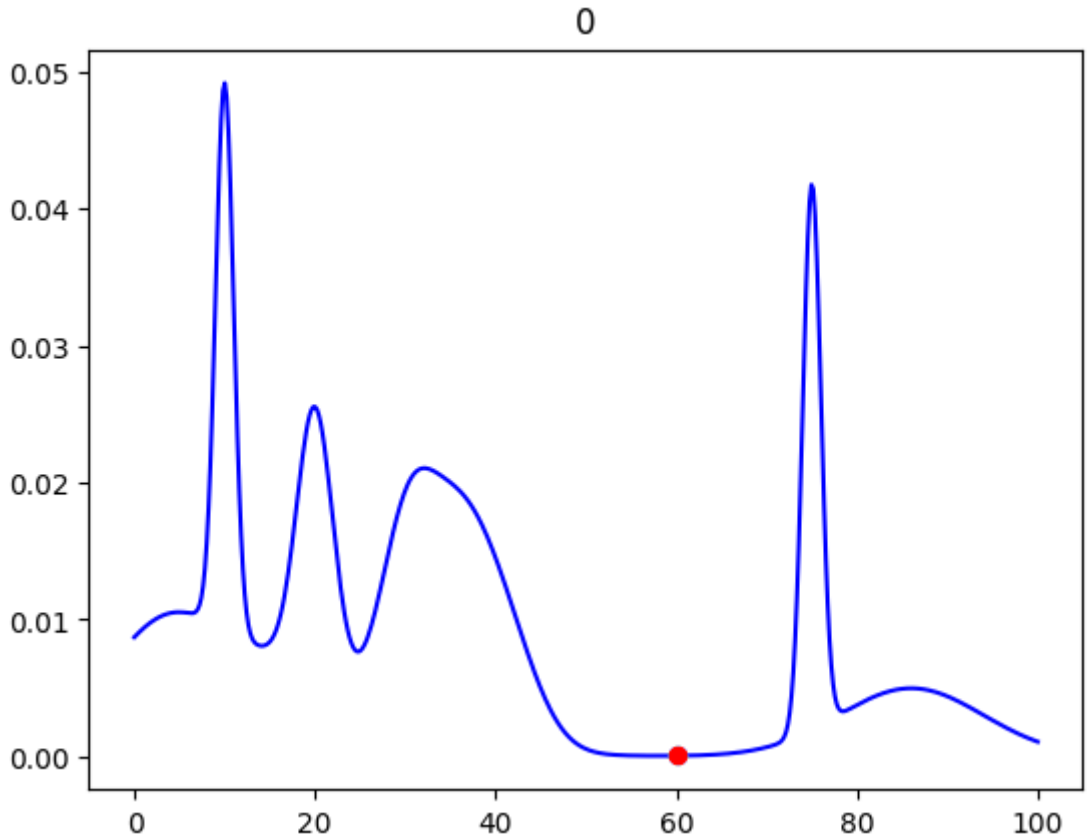
Simulated annealing

Idea: escape local maxima by allowing some "bad" moves
but gradually decrease their size and frequency

```
function SIMULATED-ANNEALING( problem, schedule) returns a solution state
    inputs: problem, a problem
            schedule, a mapping from time to "temperature"
    local variables: current, a node
                     next, a node
                     T, a "temperature" controlling prob. of downward steps

    current ← MAKE-NODE(INITIAL-STATE[problem])
    for t ← 1 to ∞ do
        T ← schedule[t]
        if T = 0 then return current
        next ← a randomly selected successor of current
        ΔE ← VALUE[next] – VALUE[current]
        if ΔE > 0 then current ← next
        else current ← next only with probability e^(Δ E/T)
```

Simulated Annealing



http://www.mad-lab.org/presentations/